

DRISHYAM

From Footage to Action

AI Requirements Document

Real-time Detection Pipeline Specification

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1. Purpose of AI in Drishyam

AI continuously analyzes live video to detect security-relevant events that humans cannot reliably monitor at scale. The output is not just a label but a verified, actionable alert with evidence.

2. Detection Classes (MVP)

Weapon: Firearm (handgun, rifle) and bladed weapons (knife, machete) held or brandished by a person.

Violence / Fight: Physical altercation between two or more people involving punches, kicks, grappling, or weapon use.

Intrusion: Person entering a restricted polygonal zone (after hours or always-restricted).

Loitering: Person remaining in a defined zone longer than a configurable threshold (e.g., 60–120 seconds).

3. Input / Output Contract

Input: Continuous video frames (or short segments) from RTSP stream, typically analyzed at 5–15 FPS and 640–1280 px width depending on edge capacity.

Output (per confirmed event):

- alert_type
- confidence (0–1)
- bounding boxes / tracks
- start and end timestamps
- optional keypoints or attributes
- reference to generated evidence clip

4. Model Strategy

- Primary detector: YOLO-family (YOLOv8/v10/v11 or equivalent) fine-tuned for weapon and person detection.
- Pose / action: Lightweight pose estimator or action recognition head for violence.
- Tracking: ByteTrack or BoT-SORT for temporal consistency.
- Deployment format: ONNX / TensorRT / OpenVINO for edge performance.
- Fallback: Cloud GPU inference for complex cases or when edge is offline (optional).

5. Temporal & Business Logic Filters

- Minimum consecutive detections (e.g., 5–10 frames) before confirming.
- Minimum duration for loitering.
- Zone membership check for intrusion.
- Confidence threshold per class (configurable).
- Cooldown / de-duplication window to avoid alert storms for the same incident.

6. Privacy Controls

- Optional face blurring on stored clips (OpenCV or model-based).
- Prefer on-edge processing so raw video does not leave the premises unnecessarily.
- No permanent facial identity database in MVP unless explicitly required and consented.

7. Fallback & Error Handling

- If model fails on a frame → skip and log; do not crash pipeline.
- If edge node is offline → camera marked degraded; optional cloud fallback if configured.
- If confidence is borderline → still create alert but flag as “low confidence” for operator review.
- Always preserve the evidence clip even if later marked false positive.

8. Performance Targets

- Inference latency on edge (Jetson-class): $\leq 100\text{--}200$ ms per frame at chosen resolution.
- End-to-end (frame to alert record): ≤ 5 seconds for 95th percentile.
- False positive rate target in pilot: $< 10\%$ after tuning.
- False negative rate for Critical classes (weapon/violence): as low as practically achievable; prioritize recall for safety.